

# Zhuoli Ouyang

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## Education

**Southern University of Science and Technology**

*Undergraduate, Information Engineering*

GPA 3.88/4.00 Rank 6/48 TOEFL Total 5.0

**Shenzhen, China**

2023 to 2027 (expected)

- **Fields Elite Class (Honors)**, Department of Mathematics
- **Honors Class**, Department of Electronic & Electrical Engineering

2023 to 2024

2024 to Present

**Dartmouth College**

*Visiting Student*

Advisor [Yaoqing Yang](#)

**Hanover, NH**

Mar 2025 to Sep 2025

**Westlake University**

*Visiting Student*

Advisor [Tailin Wu](#)

**Hangzhou, China**

Oct 2025 to Aug 2026

## Research Experience

**ENGINE Endogenous Variational MultiScale Optimization for Zeroth-Order LLM**

*NeurIPS 2026*

**Fine-Tuning**

*Accept (Poster)*

**Zhuoli Ouyang (Project Leader)**, Changxi Chi, Siyuan Li, Tailin Wu

*Advisor: Prof. Tailin Wu*

**Abstract:** Existing zeroth-order optimization (ZOO) methods for LLM fine-tuning face a trade-off between gradient-estimation variance and curvature information. We propose ENGINE, a memory-efficient ZOO framework for LLM fine-tuning. Inspired by Variational Multiscale (VMS) in computational mechanics, we treat the LoRA space as coarse and the full parameter space as fine. A V-cycle couples the two scales, using full-space curvature feedback to correct low-dimensional updates. We develop a theoretical analysis explaining how ENGINE reduces both variance and bias. ENGINE achieves SOTA performance among ZOO methods on diverse LLM fine-tuning benchmarks, with faster convergence.

### My Contributions:

- Formulated ENGINE by extending Variational Multiscale (VMS) theory from computational mechanics to ZOO. Developed the theoretical analysis explaining how coarse & fine coupling reduces both variance and bias.
- Designed the V-cycle algorithm with bidirectional state transfer from the theoretical derivations. Built and organized the complete ENGINE implementation, baselines, and evaluation framework.
- Evaluated ENGINE across model scales and downstream tasks. Shaped the paper's narrative around variance and bias reduction. Wrote the full manuscript and handled the complete rebuttal process.

**RMNP Row-Momentum Normalized Preconditioning for Scalable Matrix-Based**

*ICML 2026*

**Optimization** [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)

*Accept (Regular)*

Shenyang Deng\*, **Zhuoli Ouyang\***, Tianyu Pang, et al. *\*Equal contribution*

*Advisor: Prof. Yaoqing Yang*

**Abstract:** Muon is widely used for LLM pretraining, but its Newton-Schulz iterations incur substantial computational overhead. We propose RMNP, an efficient row-normalized optimizer motivated by prior findings on block-diagonal dominance in Transformer Hessians. Our observations further support diagonal dominance in momentum Gram matrices. Under this assumption, we derive RMNP's preconditioner as a diagonal approximation to Muon's. The two become asymptotically equivalent in the diagonal limit. RMNP replaces Newton-Schulz iterations with row-wise momentum normalization, reducing preconditioning complexity to  $O(mn)$ . It substantially reduces preconditioning time while maintaining competitive pretraining performance with Muon.

### My Contributions:

- Implemented and organized the complete RMNP, baseline, and benchmark codebase.
- Designed and ran all experiments, including benchmark comparisons and systematic ablation studies.
- Analyzed and organized experimental results. Prepared figures, tables, and experimental sections of the paper.
- Participated throughout the rebuttal. Prepared reviewer responses grounded in experimental and theoretical analysis.

**NOVA Nullspace Dynamics in Neural Basis Solvers for Moving Boundary**

*ICLR 2027 (Under Review)*

**PDEs, Transport, Rank Transitions, and Reprojection**

**Zhuoli Ouyang (Project Leader)**, Changxi Chi, Siyuan Li, Tailin Wu

*Advisor: Prof. Tailin Wu*

**Abstract:** Enforcing moving boundary constraints is challenging in multiphysics coupling. We propose NOVA to substantially reduce boundary errors while maintaining high interior accuracy. NOVA decomposes neural basis coefficients into a particular solution and a nullspace component. This enforces hard boundary constraints without penalty tuning. To reduce full SVD costs, we derive a horizontal nullspace transport equation. We justify this equation using the Stiefel principal

bundle over the Grassmannian. Based on these developments, we design an efficient algorithm. NOVA achieves SOTA boundary accuracy while maintaining high interior accuracy on the evaluated multiphysics benchmarks.

### My Contributions:

- Designed the NOVA framework to enforce hard boundary constraints through nullspace decomposition. Independently completed all theoretical derivations, establishing the mathematical foundation for nullspace transport.
- Translated the derivations into a transport algorithm to reduce full SVD costs. Implemented and organized NOVA, its baselines, and all experimental code for moving-boundary PDE evaluation.
- Designed all benchmarks, baseline comparisons, and ablations. Ran every experiment and analyzed boundary and interior accuracy. Wrote the full manuscript connecting the formulation, transport theory, and numerical results.

### OrthVMS Orthogonal Variational Multiscale Correction for Full-Rank Updates with Low-Rank Optimizer States

**Zhuoli Ouyang (Project Leader)**, Yujun He, Hongbin Zhong, et al.

*Advisor: Prof. Zhihai He*

**Abstract:** LLM training requires substantial GPU memory. Existing low-rank optimizers save memory but underuse gradient information in the orthogonal complement. We propose OrthVMS, a memory-efficient optimizer. Inspired by Variational Multiscale (VMS) in computational mechanics, we treat the projected space as coarse and its orthogonal complement as fine. We construct a full-space preconditioner from decoupled coarse and fine preconditioners. OrthVMS achieves SOTA performance among memory-efficient optimizers on diverse pretraining benchmarks.

### My Contributions:

- Proposed the core idea of OrthVMS by applying Variational Multiscale (VMS) theory from computational mechanics to LLM optimization. Constructed a full-space preconditioner from coarse and fine components.
- Established the complete convergence analysis as the paper's theoretical foundation. Designed the novel algorithm from these derivations. Implemented and organized OrthVMS, baselines, and the evaluation framework.
- Designed pretraining experiments with progressively increasing sequence lengths. Developed the manuscript's storyline linking optimizer design, convergence theory, and empirical evidence. Wrote the full paper.

## Publications

*NOVA Nullspace Dynamics in Neural Basis Solvers for Moving Boundary PDEs, Transport, Rank Transitions, and Reprojection*

**Zhuoli Ouyang**, Changxi Chi, Siyuan Li, Tailin Wu

*ICLR 2027 (Under Review)*

*OrthVMS Orthogonal Variational Multiscale Correction for Full-Rank Updates with Low-Rank Optimizer States*

**Zhuoli Ouyang**, Yujun He, Hongbin Zhong, et al.

*ICLR 2027 (Under Review)*

*ENGINE Endogenous Variational MultiScale Optimization for Zeroth-Order LLM Fine-Tuning*

**Zhuoli Ouyang**, Changxi Chi, Siyuan Li, Tailin Wu

*NeurIPS 2026 (Poster)*

*RMNP Row-Momentum Normalized Preconditioning for Scalable Matrix-Based Optimization*

Shenyang Deng\*, **Zhuoli Ouyang\***, Tianyu Pang, et al. \*Equal contribution

*ICML 2026 (Regular)*

*Understanding Generalization from Embedding Dimension and Distributional Convergence*

Junjie Yu\*, **Zhuoli Ouyang\***, Haotian Deng\*, et al. \*Equal contribution

*ICML 2026 (Regular)*

*Depth, Not Data An Analysis of Hessian Spectral Bifurcation*

Shenyang Deng\*, Boyao Liao\*, **Zhuoli Ouyang\***, et al. \*Equal contribution

*ISIT 2026*

*Suspicious Alignment of SGD A Fine-Grained Step Size Condition Analysis*

Shenyang Deng, Boyao Liao, **Zhuoli Ouyang**, et al.

*ALT 2026*

[Full publication list on my homepage](#)

## Honors & Awards

**2026** ALT 2026 Best Student Paper Award

**2026** ISIT 2026 Strong Accept

**2026** DATE 2026 Oral Paper

**2024** SUSTech, Department of Electronics and Electrical Engineering, Honor Class

**2023** SUSTech, Mathematics Department, Fields Elite Class (Top 1%, Honor Class)

## Skills

### Programming

Python, C, C++, MATLAB, LaTeX, etc.

### Tools & Frameworks

PyTorch, PETSc, Linux, Git, Docker, MATLAB, Fluent

### AI Tools

Proficient in using Claude Code, Codex, and AI agents for literature review, method exploration, code implementation, experiment execution, and result analysis.